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Dynamic graph neural network-based fraud detectors against collaborative fraudsters

Lingfei Ren [a,b], Ruimin Hu [a,b,e,*], Dengshi Li [a,c], Yang Liu [d], Junhang Wu [a,b], Yilong Zang [a,b], Wenyi Hu [a,b]

[a] National Engineering Research Center for Multimedia Software, School of Computer Science, Wuhan University, Whuhan, 430072, China
[b] Hubei Key Laboratory of Multimedia and Network Communication Engineering, Wuhan University, Whuhan, 430072, China
[c] School of Artificial Intelligence, Jianghan University, Wuhan, 430056, China
[d] Institute of Computing Technology, Chinese Academy of Sciences, Beijing, 100086, China
[e] School of Cyber Engineering, Xidian University, Xi'an, 710071, China

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ABSTRACT

Telecom fraud detection is a challenging task since the fact that fraudulent behaviors are hidden in the vast amount of telecom records. More concerning, the ongoing coronavirus pandemic (COVID-19) accelerated the use of mobile internet, providing more criminal opportunities for fraudsters. However, current telecom fraud detection mostly focuses on individual sequences representation, rarely noticing the collaboration of fraudsters, making it exhibit unsatisfactory performance in the face of gang crimes. To address this problem, we propose to extract collaborative networks from user call logs with an emphasis on unveiling collaborative fraud. We employ eight months of telecom datasets in China with 6,106 users and 5.0 million call logs between 1.25 million telephone recipients. Through our study, we find that the social structure of fraudsters evolute rapidly while the normal users remain stable relatively. In addition, we find that mining collaborative fraud strategies help to detect fraudsters with less distinct fraud characteristics. To this end, we propose a novel model named COllaborative-REsistant Dynamic Graph Neural Network (CORE-DGNN), to enhance the dynamic GNN aggregation process. Specifically, we first use co-recipients to obtain the collaborative network under each time slice. Then, we design a multi-frequency graph neural network to adaptively aggregate the features of node neighbors at different frequencies to address the problem of heterophily in collaborative networks. Finally, a self-attentive temporal convolutional network is designed to aggregate node embedding features across multiple time spans. Comprehensive experiments on two real-world telecom fraud datasets show that our approach outperforms several state-of-the-art algorithms.

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1. Introduction

The surface area of telecom fraudulent activities has increased manifold with the development of global communication technology, resulting in millions of people being exposed to fraud terribly [1]. More concerning, the coronavirus pandemic (COVID-19) has brought an unexpected sudden shock to people's lives, resulting in a decrease in offline communication and accelerated use of mobile networks, which raises new challenges in effective telecom fraud detection [2]. For example, from March 2020 to April 2021, UK police received more than 6000 cases of epidemic telecom-related fraud, with a total of £34.5 million loss. Meanwhile, Google reported that they blocked more than 240 million COVID-themed spam emails every day. Therefore, telecom fraud detection has become an essential task for both mobile service providers and governments.

To detect telecom fraud, manual feature-based models [3–5] and static graph-based models [1,4,6,7] have become effective approaches in both academic and industrial communities. For example, Diane et al. [5] proposed a threshold alarm telecom fraud identification method based on user-personalized signature features. Yang et al. [1] proposed a probabilistic factor graph to mine fraudsters and fraudulent strategies in large-scale mobile social networks. Nevertheless, telecom fraud is dynamic. On the one hand, the fraudulent behavior of the same telecom fraudster evolves to meet new fraudulent tasks. On the other hand, the dynamic evolution of telecom fraud is often quite different from normal telephone behavior. The above two static approaches only focus on the static characteristics of telecom fraud, but ignore the dynamic characteristics of telecom fraudulent behavior, resulting in poor performance.

* Corresponding author at: National Engineering Research Center for Multimedia Software, School of Computer Science, Wuhan University, Whuhan, 430072, China.
E-mail address: hrm@whu.edu.cn (R. Hu).

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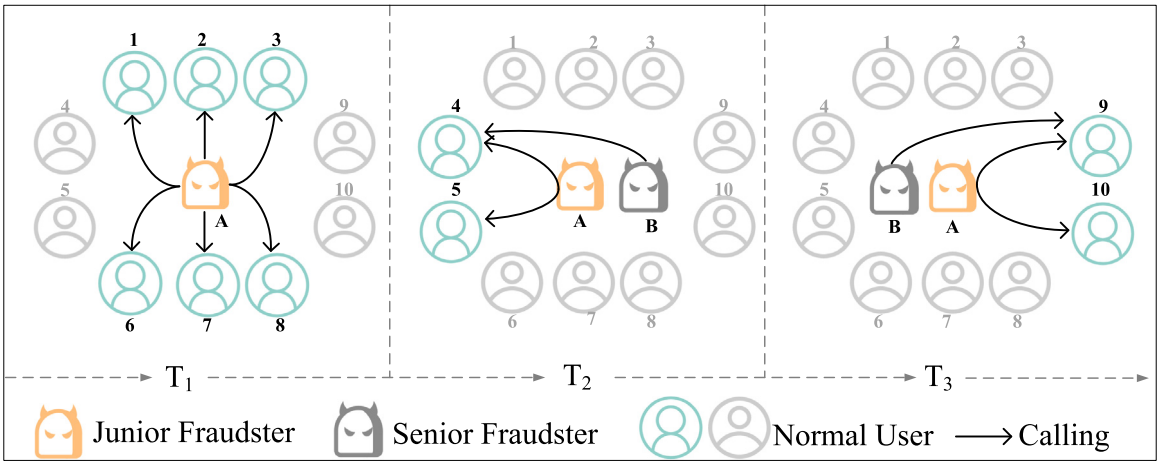


Fig. 1. An example of collaborative fraud. Due to differences in the cooperative division of labor, there are significant differences in the telephone social behavior of different fraudsters. Junior fraudster "A" tends to have a distinct telecom fraudulent pattern due to making a large number of calls to each person with high frequency to find as many potential victims as possible. However, senior fraudster "B" has a vague telecom fraud pattern due to making a small number of calls only when potential victims "4" and "9" are hooked.

Recently, inspired by the excellent performance of recurrent neural network (RNN) (e.g., LSTM) in dealing with behavioral sequence modeling [8], many RNN-based fraud detection models have been introduced to discover abnormal sequence patterns of telecom fraudsters [9–11]. These studies mainly reconstruct normal sequential patterns to predict future behaviors, and the sequences with large prediction losses tend to be identified as fraud. In contrast to traditional manual feature-based models and static graph-based models, the RNN-based method provides a better representation of fraudsters by mining dynamic features. However, existing RNN-based fraud detection methods are generally based on the assumption that the individual behavior sequences of telecom fraudsters are significantly different compared to normal users, ignoring the influence brought by collaborative fraud, resulting in the inability to correctly detect fraudsters with less distinct fraud characteristics.

To demonstrate the challenges posed by collaborative fraud, we simulate the collaborative fraud process of two fraudsters (junior fraudster "A" and senior fraudster "B") in a fraud gang. As shown in Fig. 1, the telephone social structure of junior fraudster "A" changes dramatically in the process of constantly making massive calls to find as many potential victims as possible, his telecom sequential pattern is significantly different from normal users. However, the senior fraudster "B" who is jointly a member of the same fraud gang as fraudster "A", only makes a small number of calls to induce the chosen preys to transfer money when victims "4" and "9" are hooked, having a vague fraud sequential pattern, his telecom sequence pattern is similar to that of a normal user. Obviously, the existing RNN-based fraud detection methods can easily identify fraudster "A" as a fraudster while misidentifying fraudster "B" as a normal user.

In this work, we study a large-scale mobile social network collected in a Chinese province, which covers the complete call records of 1962 telecom fraudsters and 4144 normal users over an 8-month period, and find several notable characteristics in distinguishing fraudsters from normal users from the observing the individual behaviors and collaborative behaviors. From the perspective of individual behavior, we find that the behavior of fraudsters and normal users exhibit significant differences. The behavior of fraudsters changes significantly over time, while the behavior of normal users remains stable. Concretely, fraudsters make a large number of calls to find new victims leading to rapid changes in social structure, in contrast to normal users who maintain long-term contact with family and friends leading to a stable social structure. From the perspective of collaborative behavior, we find that the current telecom fraud is mainly a gang-based crime, and there are significant differences in the behavior of different fraudsters. Specifically, certain fraudsters make a large number of fraudulent calls, and their telecom fraud characteristics are obvious, while other fraudsters have only a few calls recorded and their telecom fraud characteristics are ambiguous. We further find that combining the behaviors of users who share common recipients would be beneficial and mitigate the influence caused by collaborative fraud.

Therefore, the observed individual dynamic behavior and the collaboration between fraudsters can serve as important clues for distinguishing fraud, which, however, have merely been considered in prior work. In this work, we design a novel telecom fraud detection model named COllaborative-REsistant Dynamic Graph Neural Network (CORE-DGNN) to dynamically mine individual behaviors and collaborative behaviors. In particular, we divide all call records into several time slices with a fixed interval and build collaborative networks using common recipients within each timeslot. Also, a frequency filtering-based embedding method is proposed to aggregate the neighbor features of nodes under each time slice. In addition, a self-attention temporal convolutional network with max pooling is proposed to aggregate user representations across time slices.

It is worthwhile to highlight our contributions as follows:

- Based on real telephone communication data, we reveal the difference in telephone behavior evolution between fraudsters and normal users, i.e., fraudsters' telephone behavior changes rapidly, while normal users remain relatively stable. At the same time, we further reveal the phenomenon of collaborative fraud, and find that there are differences in the division of labor among members of the same gang, fraudsters tend to collaborate to defraud the same target.
- We propose a novel framework for telecom fraud detection. By building collaborative networks, we achieve more accurate fraud detection through the dynamic mining of individual and collaborative behaviors.
- Experiments are conducted on two large-scale mobile networks in real world and the experimental results validate the effectiveness of our model.

The rest of our paper is organized as follows. Section 2 describes a comprehensive survey of related work. Section 3 describes the motivating analysis. Section 4 details the problem

statement and the model design of our method. Section 5 presents the experimental setup and the corresponding results. Finally, in Section 6 we summarize the paper and discuss future work.

2. Related work

Due to the serious damage caused by telecom fraud, how to design efficient telecom fraud detection methods has attracted a tremendous amount of research work. In this section, we review the related research in the following aspects.

2.1. Classifier-based methods

Telecom fraud detection is formulated as a binary classification problem, i.e., given a set of telephone accounts and corresponding telecom records, predict whether an account is normal or a fraudster. Dominik et al. [3] proposed a probabilistic model for telecom network fraud detection to solve the sample imbalance problem in fraud detection. Murynets et al. [4] proposed three classifiers, alternating decision tree, function tree, and random forest, to jointly learn the telephone fraud behavior patterns. However, the classifier-based approach suffers from the effects of label sparsity, which can easily lead to the overfitting problem. In this work, we propose a semi-supervised learning framework to further exploit the unknown labels to improve performance.

2.2. Static graph-based methods

Inspired by graph neural networks (GNNs) that have shown great potential in the field of temporal knowledge graph extrapolation [12] and knowledge tracing [13,14], static graph-based fraud detection [15–17] has been the focus of recent attention. In its core sense, the underlying assumption of graph-based fraud detection is that users and fraudsters have rich behavioral interactions during telephone interactions or posting reviews, and such interactions can be represented as graph-like data that provide effective multi-dimensional information for fraud detection. For example, Chen et al. [18] proposed an automated solution for fraud detection based on graph learning algorithms to defend against insurance fraud in the e-commerce platform. Dou et al. [19] formulated a game theoretical model and reinforcement learning algorithm to find a robust detection strategy against online reviews fraud. In recent years, many researchers have focused on the class imbalance problem [15,17,20] and the camouflage problem [21–23] faced by graph-based fraud detection. However, the static graph-based fraud detection method assumes that the node characteristics and social structure are static, which often loses important dynamic information. On the one hand, the fraudster looks for more fraudulent objects resulting in "fast-varying" in a social network; On the other hand, the fraudster's fraudulent strategies are constantly upgraded during the confrontation with the fraud detection system. Therefore, this paper proposes a fraud detection method based on dynamic mining of such time-evolving differences.

2.3. Sequence-based methods

Sequence-based methods generally identify a user's behavioral sequence pattern, which is anomalous if it deviates significantly from the normal sequence. Prior research focused on sequence similarity-based techniques [24] or Markov model-based techniques [25]. More recently, deep learning methods based on sequence data have become another important class of fraud detection methods, in which the representative research is based on recurrent neural networks (RNNs). In particular, Branco et al. [11] presented an RNN framework to detect fraud in real-time. Zhu et al. [26] proposed a hierarchical interpretable network (HEN) to model the user's behavior sequences, improving the performance of fraud detection and making the inference process interpretable. Jiang et al. [27] further investigate the difference between source and target neighbor sequence patterns of normal and telecom fraudsters, and propose HESM for telecom fraud detection by integrating the Hawkes process into LSTM. Unlike telecom fraud detection based on sequence patterns that focus only on modeling individual behavior sequences, we further investigate the gang crime phenomenon and find that collaborative fraud strategy will result in less distinctive individual fraud patterns. To this end, we further propose a telecom fraud detection approach based on the dynamic mining of collaborative fraud patterns.

3. Motivating analysis

We have visualized and statistically analyzed the telecom records to demonstrate the large differences between telecom fraudsters and normal users in terms of individual behavior and collaborative behavior, as detailed below in two parts.

3.1. Real-world telecom dataset description

We obtain two telecom fraud detection datasets of different sizes from a mobile operator in one Chinese province over different periods. One dataset (i.e., Tele-max) covers 5.0 million telecom records between 1962 telecom fraudsters and 4144 normal users with 1.23 million recipients, spanning a period of 8 months from August 2019 to March 2020. The other dataset (i.e., Tele-min) covers 270.0 thousand telecom records between 607 telecom fraudsters and 1438 normal users with 0.07 million recipients, spanning the period from April 1, 2020, to April 30, 2020. Notably, the two datasets were collected randomly and only one of the datasets existed for any of the labeled users. All personal information was anonymized to protect privacy. Each user has personal information including the number of telephone cards held and monthly bills while each telecom log includes the telephone ID, call type, call start time, and call duration. We also obtain a label as to whether the user is a fraudster. In the whole paper, we only reported the overall statistical data of this dataset, without revealing any identifiable personal information.

3.2. Distinguish fraudsters from others

To understand how fraudsters differ from normal users, we examine a wide range of dynamic features over time, which include outdegree, indegree, indegree/outdegree ratio, monthly bill, call duration, and call time in different months. Specifically, we show the results for the Tele-max dataset. Meanwhile, similar results can be obtained for Tele-min.

Fig. 2(a) and (b) show the trends of users' outdegree and indegree over time, respectively. As expected, the outdegree and indegree of normal users are stable, while the fraudsters fluctuate over time. This suggests that fraud behavior varies significantly over time. In addition, we are surprised to find that the outdegree of fraudsters increases dramatically from October 2019 to January 2020, which we speculate is due to the fact that January 2020 is the Chinese Spring Festival and telecom fraudsters are likely to adopt more aggressive fraudulent methods during important holidays. Therefore, we should be more careful once receive calls from strangers during important holidays, as the callers are likely to be fraudsters.

Fig. 2(b) shows that fraudsters receive higher callbacks than normal users. In particular, from November 2019 to January 2020,

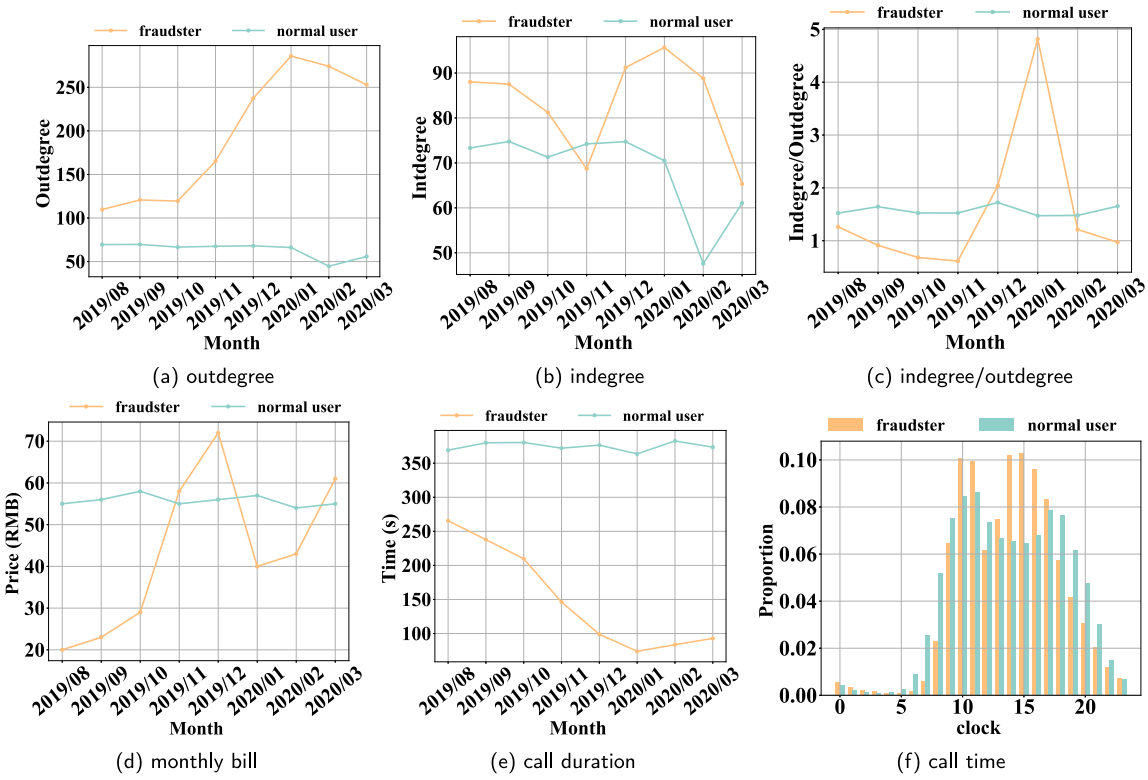


Fig. 2. Dynamic feature comparison between fraudsters and normal users. The telephone behaviors of fraudsters are different from those of normal users: the behaviors of fraudsters change significantly at different times while normal users keep stable.

there is a sharp increase in the indegree of fraudsters, which indicates to some extent that fraudster is more likely to succeed during important holidays. In addition, from Fig. 2(c), we can see that the indegree and outdegree ratios of fraudsters change dramatically over time.

From Fig. 2(d), we can see the monthly telephone bills of normal users are relatively stable, while the fraudsters vary considerably. This proves that the number of calls made by fraudsters varies significantly. Specially, in August 2019, the fraudster's telephone bill was only 22.25, indicating that other methods of fraud may have been used during this period. However, in December 2019, the fraudster's telephone bill was as high as 72.64, indicating that the fraudster conducted fraud primarily by telephone during this period.

Based on users' telecom records, we define the call duration dispersion, which is the average telephone duration of users with each recipient. As shown in Fig. 2(e), the average telephone duration of fraudsters with each recipient is constantly lower than that of normal users, which indicates that fraudsters are not willing to waste much time with fixed recipients and keep looking for new targets. Normal users, on the other hand, prefer to maintain steady contact with regular recipients.

Fig. 2(f) shows the distribution of calls made by users at different times of the day. We can find that fraudsters make more calls during working hours (e.g., 10 a.m. to 5 p.m.) and fewer calls during non-working hours (e.g., 5 p.m. to 10 a.m.) compared to normal users. This suggests that fraudsters work like office employees with labor discipline.

3.3. Disclose collaborations between fraudsters

We use visualization and statistics to demonstrate the presence of collaborative fraud in telephone networks and show that aggregating the fraudster characteristics with collaboration helps to distinguish fraudsters from normal users.

Do fraudsters adopt collaborative strategy? We assume that there is collaborative fraud among fraudsters with more than a threshold number of common recipients. In this paper, we set the threshold to 2, i.e., fraudsters with more than 2 common recipients adopt collaborative fraud strategies. As shown in Fig. 3(a), more than 1000 telecom fraudsters in the dataset adopt collaborative fraud strategies, while less than 800 telecom fraudsters consistently adopt individual fraud strategies.

We further explore the impact of collaborative fraudulent behavior. We counted the number of fraudulent phone calls, and as shown in Fig. 3(b), about 25% of fraudsters made less than 50 calls in the past 8 months. We speculate that these few callers are more likely to be senior fraudsters who generally only form coordinated fraud teams to lure money when their victims are hooked by junior fraudsters. Obviously, these senior fraudsters tend to be perceived as normal users and escape detection

How to defend against collaborative fraud? We further leverage common call recipients to build collaborative networks and aggregate the individual feature of neighbors as node collaborative features. Fig. 3(c) visualizes the average intra-similarity (average cosine similarity between fraudster and fraudster) and average inter-similarity (average cosine similarity between fraudster and normal user) of individual fraud and collaborative fraud. We can see that the average intra-similarity of collaborative fraud is higher than that of individual fraud, while the average inter-similarity of collaborative fraud is lower than that of individual fraud, which shows that aggregating neighbors features in collaborative networks can improve the separability of fraudsters and normal users.

4. Proposed methodology

4.1. Problem formulation

Let $U = \{u_1, u_2, \ldots, u_N\}$ be the set of N users and $V = \{v_1, v_2, \ldots, v_L\}$ be the set of L recipients.

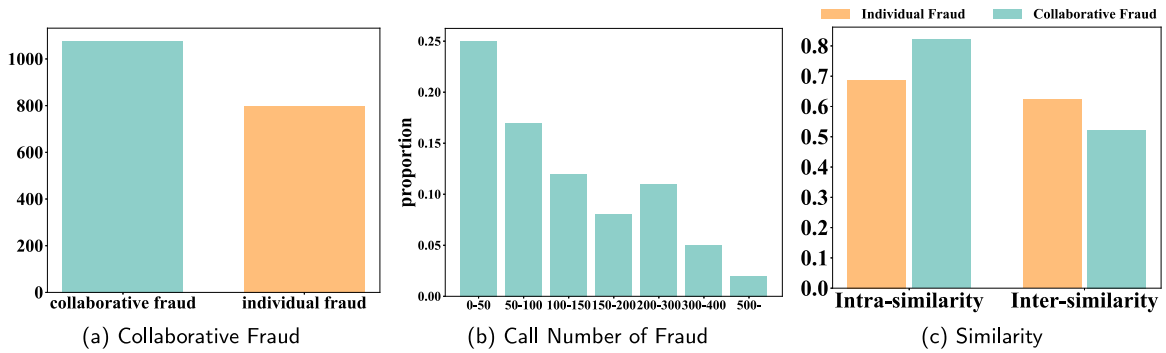


Fig. 3. Collaborations between fraudsters. (a) Evidence of collaborative fraud. More than 1000 telecom fraudsters adopt collaborative fraud strategies, while less than 800 telecom fraudsters consistently adopt individual fraud strategies. (b) Call number of fraud, where about 25% of fraudsters made less than 50 calls in the past 8 months. (c) The intra-similarity and inter-similarity of individual fraud and collaborative fraud.

Table 1
Features that used in fraud mining.

Attributes name	Description
Static feature of user u_i	
Telephone cards number	The number of telephone cards held by u_i
Monthly bills	The monthly telephone bills of u_i, which varies from 2019.08 to 2020.03
Call behavior of user u_i	
Outdegree d^i_{out}	The number of calls made by u_i
Indegree d^i_{in}	The number of callbacks received by u_i
Indegree/Outdegree ratio r^i	The ratio of u_i's indegree to outdegree d^i_{in}/d^i_{out}
Neighbors number n^i_{neigh}	The number of recipients that received call from u_i
Neighbor degree d_{neigh}	The average calls number made by u_i to each recipient n^i_{neigh}/d^i_{out}
Neighbor repetition period p^i_{neigh}	The average repetition period of recipient called by u_i
Energy distribution $dist^i$	The average call during with each recipient of u_i
Call interval t^i_{gap}	The average time interval between two calls made by u_i
Call during t^i_{dur}	The average during of calls by u_i
Call time t^i_{call}	The average time of calls by u_i, which varies from 0:00 am to 23:59 pm.

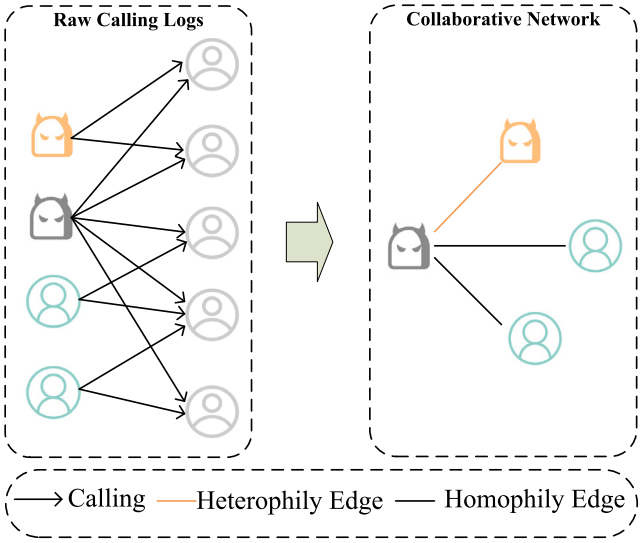


Fig. 4. Heterophily problem in collaborative network.

Definition 1 (*Call Log*)**.** When a user u calls with a recipient v at time t, the call log can be modeled as a quadruple $r^u_t = \langle u, v, c, t\rangle$, where c denotes the call characteristics which contains call type and during.

Definition 2 (*Call Sequence*)**.** Given a user u, the call sequence is a set of chronologically ordered call logs denoted as $R_u = \langle r^u_{t_1}, r^u_{t_2}, \ldots, r^u_{t_K}\rangle$, where K denotes the call sequence length.

Definition 3 (*Homophily Edge and Heterophily Edge*)**.** For a graph, if two nodes connected by an edge belong to the same class of labels, then this edge is a homophily edge. Otherwise, this edge is heterophily edge.

Definition 4 (*Telecom Fraud Detection*)**.** Formally, given all users U with call sequences $R_U = \langle R_{u_1}, R_{u_2}, \ldots, R_{u_N}\rangle$. Each user u has a label $y_u \in \{0, 1\} \in Y$ where 0 represents normal user and 1 represents telecom fraudster. Telecom fraud detection is a semi-supervised binary node classification problem base on user call sequence. Telecom fraudster detectors are trained based on the labeled users information, the trained models are then used to predict the suspiciousness of unlabeled users.

4.2. Methodology

We achieve telecom fraud detection with a novel model, i.e, CORE-DGNN. As shown in Fig. 5, the CORE-DGNN model mainly contains two components: the multi-frequency graph neural network module to learn the collaborative behavior at any time snapshot and the self-attention temporal convolution module to learn the dynamic pattern over time in a hierarchical manner. First, we divide the raw call logs into multiple snapshots using time intervals s (e.g., 30 days). In each snapshot t, we construct a collaborative network $G_t = \{U_t, E_t, X_t\}_{1\le t\le T}$ using co-recipients threshold d (e.g. 2), where the individual features X_t are extracted as Table 1. Note that, the set of nodes U_t may change more over time, as some users may have no call behavior for some time. In our example, U_t is assumed to be fixed. However, collaborative networks constructed by manual rules inevitably contain noises [28–30], which leads to heterophily problem [21,

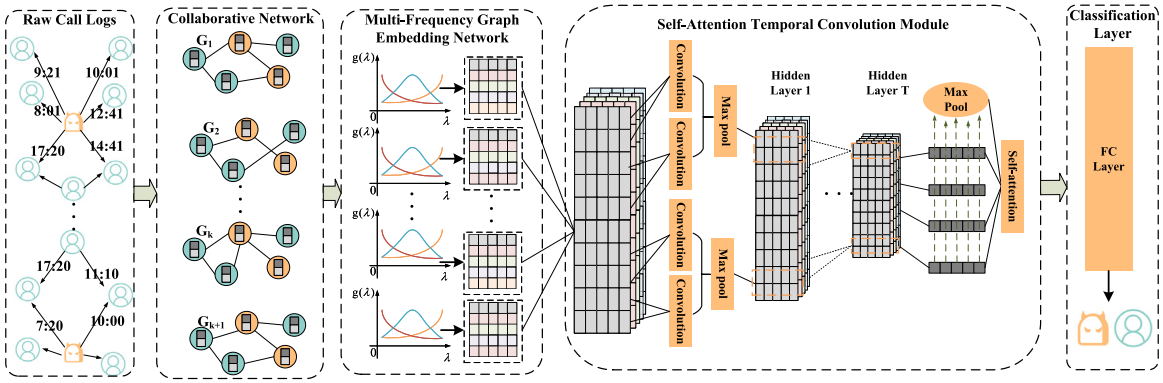


Fig. 5. Illustration of the proposed method.

31]. As shown in Fig. 4, telecom fraudsters may have multiple co-recipients with normal users in raw calling logs, leading to the establishment of connections between them in the collaborative network, and thus an increase in the heterophily edges. However, GNNs use low-pass to maintain the commonality of neighboring node features and assimilate the representations of connected nodes even if they have different labels. This aggregation strategy may introduce noise in node representation learning and reduces the performance of fraud detection [21]. Inspired by BWGNN [32], we design a multi-frequency graph neural network to aggregate the neighbors feature to the node. Specifically, the low-frequency filter aggregate the common feature of neighbors with the same label, and a high-frequency filter aggregates the disparity information of neighbors with a different label. Next, we use a self-attention temporal convolutional network to aggregate node embedding $H^i_{0\le t\le T}$ under all snapshots to obtain the final embedding Z^i.

4.3. Multi-frequency graph neural network

In order to efficiently learn collaborative behavior in collaborative networks, we use the graph neural network (GNN) to aggregate the common feature of neighbors with the same label. Its input is the graph structure and user raw features, and the output is the user embedding representation. Motivated by the success of BWGNN in the field of graph-based fraud detection, our multi-frequency graph neural network uses Beta Wavelet.

Graph Fourier Transform. According to the theory of graph signal processing [33], the normalized Laplacian matrix of the graph $\mathbf{L}$ can be decomposed into multiple orthogonal bases $\mathbf{U} = (\mathbf{u_1}, \mathbf{u_2}, \ldots, \mathbf{u_N})$, i.e. $\mathbf{L} = \mathbf{U\Lambda U}^T$, where $\mathbf{\Lambda} = \mathbf{diag}[\lambda_1, \ldots, \lambda_n]$ denotes the eigenvalue corresponding to the orthogonal base. Given a signal $x \in R^n$, its graph Fourier transform is $\widehat{x} = \mathbf{U^T}x$ and the inverse graph Fourier transform is $x = U\widehat{x}$. Thus, the signal x_t for network snapshot t passes through the filter f gives:

$$f * x_t = U_t(U_t^T f) \odot (U_t^T x_t) = U_t f(\lambda) U_t^T x_t, \tag{1}$$

to parameterize the filter, $f(\lambda)$ is usually set to a polynomial of order K with eigenvalue λ [34–36] :

$$f(\lambda) = \sum_{k=0}^{K} a_k \lambda^k \tag{2}$$

Then, the filtering process can be expressed as:

$$U_t f(\lambda) U_t^T x_t = \sum_{k=0}^{K} a_k U_t \lambda^k U_t^T x_t = \sum_{k=0}^{K} a_k L_t^K x_t \tag{3}$$

Beta Wavelet Graph Neural Network. The probability density of Beta distribution:

$$f_{p,q}(w) = \begin{cases} \frac{w^p (1-w)^q}{B(p+1,q+1)} & if\ w \in [0,1] \\ 0 & otherwise \end{cases} \tag{4}$$

where $p + q = C$ which is a hyperparameter to control the field of view of the filter and $B(p+1, q+1) = p!q!/(p+q+1)!$ are constants. For the filter function f: $[0, 2] \to [0, 1]$, we let $w = \lambda/2$. Thus, the signal x_t passing through the Beta wavelet filter $f_{p,q}(\lambda)$ gives:

$$U_t f_{p,q}(\lambda) U_t^T x_t = \frac{L_t^p (1 - L_t)^q}{B(p+1, q+1)} x_t \tag{5}$$

where the L is the normalized Laplacian matrix of graph. Different from the structure-based approaches, we obtain node embeddings H_t in snapshot t using the aggregation of multiple filters of different frequencies:

$$\begin{aligned} Z_i^t &= f_{i,C-i}(\lambda)(MLP(X_t)) \\ H_t &= AGG(Z_1^t, Z_2^t, \ldots Z_C^t) \end{aligned} \tag{6}$$

where the MLP is a multi-layer perceptron and *AGG* is concatenation.

4.4. Self-attention temporal convolution component

We propose a self-attention temporal convolution module to mine fraudsters' dynamic behavior under multiple time snapshots. Its input is a sequence of user embeddings $\{H_t\}_{1\le t\le T}$ from the multi-frequency graph neural network module obtained via Eq. (6) and the output is the embedding Z for users. However, mining dynamic fraud behavior is a non-trivial task, as fraudsters' behavior is often discontinuous but randomly scattered throughout time. In addition to this, mining sequential patterns of behavior may not be as important for determining fraudsters, but rather mining significant fraudulent behavior may be more important. Thus, using RNN/LSTM to mine dynamic fraudulent behavior is not optimal. In this paper, we design a temporal convolutional (TCN) layer with max-pooling to mine local and global fraud representations, and then, use a self-attentive mechanism to collect representations scattered across the time and assign different weights to local and global representations.

Temporal Convolutional Network. Similar to RNN/LSTM, TCN mainly mines sequential patterns [37], while TCN can flexibly change the perceptual field to suit different tasks by changing the embedding depth and the size of the convolution kernel. The TCN is based on two principles: the output is the same length as the input, and there is no leakage of time. To accomplish the

first point, the TCN uses a one-dimensional fully convolutional network with zero-length padding added to keep subsequent layers the same length as the previous ones. To accomplish the second point, the TCN uses causal convolution, where the output at time t is only convolved with time t and the previous elements. Given an input sequence $\widehat{Z}_t^{(l-1)} = \{\widehat{z}_0^{(l-1)}, \widehat{z}_1^{(l-1)}, \widehat{z}_2^{(l-1)}, \ldots, \widehat{z}_T^{(l-1)}\}$ in layer $l-1$ and a 1D fully-convolutional kernels $f_{0\le t\le T}^{(l-1)}$, the embedding in layer l can be obtained as:

$$\begin{aligned}\widehat{Z}_t^{(l)}(s) &= ReLU(\widehat{Z}_t^{l-1} * f^{(l-1)}) \\ &= ReLU(\sum_{i=0}^{k-1} f^{(l-1)}(i)\widehat{Z}_t^{(l-1)}(s-d,i)) \end{aligned} \tag{7}$$

$$\{\widehat{Z}_t^{(0)}\}_{0\le t\le T} = \{H_t\}_{0\le t\le T}$$

where d denotes the size of the convolution kernel and $*$ denotes the convolution operation and the length of $\widehat{Z}_t^{(l)}$ is T.

Temporal Convolutional Network with Max-pooling. Temporal convolution can only extract sequential patterns of fraudulent behavior. As mentioned previously, we prefer the model to extract the most significant part of the local features than the sequential patterns. For example, fraudster A camouflages as a normal person in snapshots T_1 and T_2 and takes intensive fraudulent behavior in snapshot T_3, we would like the model to focus on the user representation in T_3, rather than keeping all features from T_1, T_2 and T_3, which makes it challenging to apply the temporal convolution to fraud detection tasks. Our solution here is to use a max-pooling strategy, selecting the most distinctive local features in each layer of the temporal convolution to be retained and then passed on to the next layer, and so forth to obtain.

$$Z_t^{(l)} = Max - pool(\widehat{Z}_{2t+1}^{(l)}, \widehat{Z}_{2t}^{(l)}) \quad for \;\; 1 \le l \le L \tag{8}$$

where the length of $Z_t^{(l)}$ is $T/2^l$. Obviously, $Z_t^{(l)}$ maintains the most significant local features of the node.

Self-attention Mechanism. We linearly merge the output of max pooling into each $Z_t^{(l)}$ to let each $Z_t^{(l)}$ capture global information.

$$Z_{\max} = Max - pool(Z_t^{(L)})_{0\le t\le\{T/2^L\}} \tag{9}$$

To capture the complex relationship between local and global representations, we propose to use the attention mechanism $att(Z_0^{(L)}, Z_1^{(L)}, \ldots, Z_{\{T/2^L\}}^{(L)}, Z_{\max})$ to automatically learn the corresponding importance $(\beta_1, \beta_1, \ldots \beta_{\{T/2^L\}}, \beta_{\max})$ and fuse them for the task of fraud detection:

$$(\beta_1, \beta_1, \ldots \beta_{\{T/2^L\}}, \beta_{\max}) = att(Z_0^{(L)}, Z_1^{(L)}, \ldots, Z_{\{T/2^L\}}^{(L)}, Z_{\max}) \tag{10}$$

where $\beta_1, \beta_1, \ldots \beta_{\{T/2^L\}}, \beta_{\max} \in {}^{N\times 1}$ are the attention values of n nodes with $Z_0^{(L)}, Z_1^{(L)}, \ldots, Z_{\{T/2^L\}}^{(L)}, Z_{\max}$, respectively. Specially, the attention score for the node v_i with embedding $z_i^{(L)}$ is:

$$w_k^i = q^T \cdot \tanh(W \cdot z_k^{(L)} + b) \tag{11}$$

where W is the weight matrix, b is the bias vector, q is the shared attention vector. The final attention weights of node u_i are obtained by normalizing the attention values w_k^i with softmax function as:

$$\beta_k^i = softmax(w_k^i) = \frac{\exp(w_k^i)}{\sum_k \exp(w_k^i)} \tag{12}$$

With the learned weights as coefficients, we can fuse local embeddings and global embedding to obtain the final embedding Z as follows:

$$Z = \sum_{k=1}^{\{T/2^L\}} \beta_k Z_k^{(L)} + \beta_{\max} Z_{\max}^{(L)} \tag{13}$$

Algorithm 1 CORE-DGNN model

Input: $\mathcal{G} = G_1, G_2, ..., G_T$: set of dynamic collaborative graphs, where $G_t = (U_t, E_t, X_t)$, user labels $y_0, y_1 ... y_N$, number of snapshots T

Output: Probability y' for fraudster or normal user.

```
Initialize the parameters;
// Multi-frequency Graph Neural Network
for 0 ≤ t ≤ T do
    Obtain H_t by Eq. (6);
end for
// Temporal Convolution Component
for 0 ≤ t ≤ T do
    for 0 ≤ l ≤ L do
        Obtain Ẑ_t^(l)(s) by Eq. (7);
        Obtain Z_t^(l) by Eq. (8);
    end for
end for
// Self-attention Mechain
Obtain Z_max by Eq. (9);
Obtain Z by Eq. (13);
// Model learning
Obtain y'_i by Eq. (14);
Obtain L by Eq. (15);
Update the parameters with the Adam optimizer.
return y'_i
```

4.5. Model learning

In a semi-supervised framework, we partially know the user labels, a small fraction of which are fraudsters that are considered as positive instances, while the rest are normal users that are considered as negative instances. Therefore, We feed the user embedding Z^i obtained by the self-attention mechanism into a fully connected layer to obtain a conditional probability distribution for node classification, and then use the softmax function to normalize the result from 0 to 1:

$$y'_i = softmax(Z^i W_i + b_i) \tag{14}$$

where W_i and b_i are parameters to be learned. Thus, we obtain the predicted probability vector y'_i, which represents the probability of a user are fraudsters or not.

We use cross entropy to calculate the loss, as follows:

$$L = -\sum_{u_i \in V} [y_i \ln(y'_i) + (1-y_i)\ln(1-y'_i)] \tag{15}$$

where the y_i is the ground truth label of the user u_i.

The parameters in our model are optimized using the Adam optimizer, a variant of stochastic gradient descent. Algorithm 1 outlines the training process for the CORE-DGNN method. Our CORE-DGNN method requires the following parameters to be learned: (1) parameter matrices in the multi-frequency graph neural network module and temporal convolutional network with max-pooling module; (2) parameter matrices and bias vectors in the self-attention mechanic; (3) parameters for fusion in the classification layer.

5. Experimental setup

In this section, we show the results of a series of experiments using the dataset presented in Section 3 to evaluate the effectiveness of our proposed approach. All experiments are implemented in PyTorch on Ubuntu 20.04.1 LTS server with Cuda 11.4.

5.1. Comparison method

We employ the following methods as baselines:

Classifier-based methods are as follows:

FFD [1]: A semi-supervised static probability graph model to solve the telecom fraud detection in large-scale telephone social networks, which introduces the telephone social characteristics and call behavior (e.g. call duration) of the fraud users.

FRAUDAR [38]: A graph-based unsupervised algorithm supporting bipartite graphs, so we modify the telephone social network by defining two sets: the user who makes the call (known label) and the user who receives the call (no label). The social network is created from raw call logs.

Static graph-based methods are as follows:

GCN [39]: A graph convolution network achieved by aggregating local first-order neighbor nodes.

GAT [40]: A graph attention network that employs an attention mechanism for neighbor aggregation.

CARE-GNN [22]: A camouflage-resistant graph neural network for review fraud detection which enhances the GNN aggregation process with three unique modules against camouflages.

PC-GNN [15]: A GNN-based imbalanced learning method to solve the class imbalance problem in review fraud detection. Especially, the center nodes are picked with a label-balanced sampler to construct a balanced sub-graph for mini-batch training.

BWGNN [32]: This model leverages Beta graph wavelet to generate band-pass filters to handle the 'right-shift' phenomenon of spectral energy distributions in fraud detection.

Sequence-based methods are as follows:

HAInt-LSTM [41]: This model forms a representation of the sequence of behaviors required for telecom fraud detection. When designing the interaction module, it takes as input only the raw IDs of the source and target users and connects them with the feature vectors as the output of the model. This model focuses on mining the sequential anomaly patterns of fraudsters.

NHA-LSTM [42]: This model augments the traditional LSTM with a modified forgetting gate based on the HAInt-LSTM, where the interval is the duration between consecutive time steps, and a self-historical attention mechanism is designed to allow for long-term dependence. In addition, an enhanced network embedding approach, FraudWalk, is considered to construct embeddings for nodes in the interaction network involving higher-order interactions and special temporal constraints to reveal potential group frauds. This model focuses on mining the sequential anomaly patterns of fraudsters.

BiDyn [43]: This model uses dynamic bipartite graph representations to detect abusive behavior in retail websites and communities. Specifically, it deals with static and dynamic features by introducing recurrent networks (RNN) and graph neural networks (GNN).

HESM [27]: This model extracts source and target neighbor sequences from user telecom behavior and uses a Hawkes-enhanced model to reveal interesting correlation and repetition patterns hidden in both sequences, which are important clues for fraudster detection.

5.2. Evaluation metrics

We evaluate the detection performance with three popular metrics, i.e., Macro-Precision, Macro-Recall, and Macro-F1-score, since the macro metrics can reflect the performance of fraud detectors on imbalanced data without overly underestimating the minority (fraud) class. Specifically, Macro-Precision is the unweighted mean of the proportions of the true fraudsters in all detected fraudsters and the true normal users in all identified normal ones, which is calculated as follows:

$$\begin{aligned} Precision_{macro} &= \frac{P_a + P_n}{2} \\ &= \frac{tp_a}{2\,(tp_a + fp_a)} + \frac{tp_n}{2\,(tp_n + fp_n)} \end{aligned} \tag{16}$$

where P_a and P_n measure the precision for the fraud and normal classes, respectively. tp_a counts the true positives of detected fraudsters and fp_a counts the false positives of fraudsters, while tp_n and fp_n are those for the normal class.

Similarly, Macro-Recall indicates the unweighted average of the proportions of the detected true fraudsters in ground truth fraudsters and the detected true normal users in ground truth normal ones, which is calculated as follows:

$$\begin{aligned} Recall_{macro} &= \frac{R_a + R_n}{2} \\ &= \frac{tp_a}{2\,(tp_a + fn_a)} + \frac{tp_n}{2\,(tp_n + fn_n)} \end{aligned} \tag{17}$$

where R_a and R_n measure the recall of the fraud and normal classes, respectively. fn_a counts the false negatives of detected fraudsters and fn_n counts the false negatives of detected normal users.

F1-score is a compromise between recall and precision, and Macro-F1-score is the unweighted average of the F1-scores for the two classes (i.e., fraud and normal), which is calculated as follows:

$$\begin{aligned} F1_{macro} &= \frac{F1_a + F1_n}{2} \\ &= \frac{P_a \cdot R_a}{P_a + R_a} + \frac{P_n \cdot R_n}{P_n + R_n} \end{aligned} \tag{18}$$

To obtain reliable results, we performed each experiment 10 times and used the average values for evaluation.

5.3. Default parameters and optimization

We use the Adam optimizer to optimize the parameters of CORE-DGNN with a learning rate setting of 0.001 and a weight decay of 0.005. In Tele-Max, the time interval s is 30 days, the co-recipients threshold d is 6, and the order C is 5. In Tele-min, the time interval s is 7 days, the co-recipients threshold d is 2, and the order C is 3. For all fraud methods, we use the default hyperparameter settings based on published papers. In addition, we set the application scenarios of the baseline methods for the fairness of the experiments. Specifically, the classifier-based methods (i.e., FFD and FRAUDAR) use a network constructed from the raw call records in which the nodes in the network consist of the users and the telephone recipients, and edges are established if there is one or more call records between the nodes. Static graph-based methods (i.e., GCN, GAT, CARE-GNN, BWGNN, and BWGNN) use static collaborative networks in which the nodes in the network consist of only the target users, and edges are built if there are two or more common call recipients between the nodes. The static collaborative network does not contain dynamic information about fraudulent behavior. Sequence-based methods (i.e., HAInt-LSTM, NHA-LSTM, BiDyn and HESM) use the call sequences of each user extracted from the raw call records. The sequence-based approach does not consider collaborative fraud.

5.4. Performance comparison

For telecom datasets, obtaining supervised node labels is generally difficult, so we vary the size of the training set between 10%, 20%, 30%, and 40% of the total sample and take 33% of

Table 2
Performance of telecom fraud detection under different percentages of training data. F1 (abbr. F1-macro).

Tele-Max												
Method	10%			20%			30%			40%		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
FFD	0.6323	0.5810	0.5988	0.6546	0.6045	0.6285	0.6600	0.6099	0.6285	0.6789	0.6121	0.6101
FRAUDAR	0.5372	0.6068	0.6023	0.5423	0.5989	0.5875	0.5463	0.6134	0.6128	0.5510	0.6381	0.6098
GCN	0.6615	0.6452	0.5362	0.6557	0.6410	0.5327	0.6255	0.5589	0.3818	0.6504	0.6205	0.4953
GAT	0.6341	0.6034	0.5233	0.6212	0.6023	0.4920	0.6488	0.6322	0.4928	0.6656	0.6421	0.5028
CARE-GNN	0.8442	0.8061	0.8208	0.8637	0.8264	0.8408	0.8576	0.8282	0.8404	0.8640	0.8297	0.8434
PC-GNN	0.7587	0.7925	0.7654	0.7894	0.8182	0.7987	0.7439	0.7827	0.7415	0.7553	0.7947	0.7563
BWGNN	0.8888	0.8024	0.8179	0.8936	0.8121	0.8267	0.8946	0.8101	0.8429	0.8807	0.8291	0.8555
HAInt-LSTM	0.6803	0.5235	0.5917	0.7296	0.5844	0.6470	0.7313	0.5881	0.6519	0.7390	0.5921	0.6629
HAA-LSTM	0.7274	0.7365	0.7314	0.7515	0.7683	0.7580	0.7453	0.7714	0.7528	0.7439	0.7730	0.7508
BiDyn	0.8585	0.8090	0.8317	0.8478	0.8185	0.8369	0.8411	0.8040	0.8269	0.8288	0.7731	0.7967
HESM	0.8323	0.7977	0.8499	0.8399	0.7823	0.8487	0.8423	0.7911	0.8376	0.8423	0.7811	0.8418
CORE-DGNN	**0.9078**	**0.8265**	**0.8533**	**0.9103**	**0.8330**	**0.8591**	**0.9118**	**0.8360**	**0.8619**	**0.9129**	**0.8498**	**0.8728**
Tele-Min												
FFD	0.6829	0.6023	0.6281	0.6932	0.6212	0.6295	0.7100	0.6367	0.6325	0.7200	0.6432	0.6532
FRAUDAR	0.5674	0.6222	0.6232	0.5703	0.6319	0.6265	0.5764	0.6214	0.6448	0.5978	0.6401	0.6491
GCN	0.6901	0.6721	0.5900	0.6990	0.6790	0.6010	0.7015	0.6989	0.5820	0.7014	0.6229	0.6013
GAT	0.6891	0.6451	0.5903	0.6829	0.6533	0.5990	0.7018	0.6929	0.5918	0.7136	0.6823	0.6550
CARE-GNN	0.8589	0.8343	0.8458	0.8723	0.8465	0.8608	0.8700	0.8499	0.8724	0.8831	0.8555	0.8642
PC-GNN	0.7721	0.8135	0.7853	0.7790	0.8202	0.7922	0.7909	0.7899	0.7655	0.8023	0.8140	0.7988
BWGNN	**0.9128**	0.8534	0.8320	**0.9236**	**0.8424**	0.8407	0.9277	0.8321	0.8619	**0.9290**	0.8501	0.8754
HAInt-LSTM	0.6720	0.5309	0.5819	0.7191	0.5855	0.6310	0.7423	0.5590	0.6432	0.7500	0.5811	0.6430
HAA-LSTM	0.7355	0.7260	0.7433	0.7744	0.7732	0.7830	0.7743	0.7801	0.7833	0.7609	0.7910	0.7404
BiDyn	0.8405	0.7991	0.8233	0.8572	0.8455	0.8509	0.8523	0.8322	0.8409	0.8600	0.8324	0.8507
HESM	0.8145	0.7827	0.8532	0.8509	0.8025	0.8517	0.8600	0.7845	0.8487	0.8515	0.8122	0.8544
CORE-DGNN	0.9110	**0.8591**	**0.8423**	0.9190	0.8410	**0.8501**	**0.9292**	**0.8499**	**0.8634**	0.9200	**0.8599**	**0.8800**

the remaining data as the validation set, and then use the rest of the data as the test set. After training the model shown in Algorithm 1, we apply the trained model to the test set to obtain the prediction loss for each caller.

The experimental results of the proposed CORE-DGNN and the baselines are reported in Table 2. As we can see, our model achieves the best performances with four different training set sizes in terms of the general metrics precision, recall, and F1-score. In particular, when the training is extremely scarce, such as when the proportion is 10%, the F1-score remains above 0.85, and the precision value is above 0.91, which indicates proving both the superiority and the scalability of CORE-DGNN. Further analysis of the experiment results follows:

- FRAUDAR has the worst performance, as expected. Although FRAUDAR can characterize the user's telephone social structure utilizing a bipartite graph, it ignores the telephone behavioral characteristics, which are important for determining whether a user is a telecom fraudster.
- Compared with classifier-based methods (e.g., FFD and FRUADAR), deep neural network-based methods (i.e., static graph-based methods and sequence-based methods) outperform the former, indicating that deep learning methods are more effective in modeling the complex call patterns of users; meanwhile, FDD outperforms FRAUDARE, which is mainly because FFD models the user's call behavior characteristics in addition to the user's telephone social structure characteristics.
- Static graph-based fraud detection methods (CREA-GNN, PC-GNN, and BWGNN) outperform sequence-based methods (HAInt-LSTM, NHA-LSTM, BiDyn, and HESM). This is mainly because sequence-based detection methods focus mainly on individual fraud and ignore gang crimes. The effectiveness of sequence-based methods will decrease faced with gang crimes. In contrast, static graph-based fraud detection methods can aggregate the characteristics of neighbors belonging to potential gangs to mine gang fraud users by building collaborative networks. It can be seen that gang crime has become a prevalent fraud strategy in the field of telecom fraud detection, which is consistent with the conclusions drawn in the Introduction.
- CREA-GNN, PC-GNN, and BWGNN outperform GCN and GAT, indicating that solving the graph heterophily can improve performance. Specifically, the former can not only aggregate the neighboring features of nodes to alleviate the challenge of less distinct fraud characteristics due to collaborative fraud strategies but also filter the neighbors who do not belong to the same class to reduce the degradation of performance due to inappropriate aggregation. BWGNN outperforms CREA-GNN and PC-GNN, indicating that the graph filtering approach is more suitable for graphs heterophily than the neighbor filtering approach.
- In particular, we note that in Tele-Min, the performance of BWGNN is similar to that of CORE-GNN, and even slightly better than the latter in some cases. This may be because the period of Tele-GNN is small and the dynamic behavioral characteristics of users are not obvious. Therefore, we believe that the dynamic collaborative network is more suitable for telecom fraud detection tasks in scenarios with larger periods, in which the dynamic character of fraud is evident.
- CORE-DGNN outperforms all other methods, indicating that it can capture more useful information to better mine complex fraud characteristics. The advantages of CORE-DGNN are as follows: (1) It further considers the phenomenon of collaborative fraud. By building a collaborative network based on co-recipients and aggregating the neighboring features of users to alleviate the problem of less distinct fraud characteristics caused by collaborative fraud. (2) It designs a temporal convolutional network with max-pooling that not only mines the dynamic behavioral features but also focuses on the most obvious fraudulent behavior while filtering out non-anomalous behaviors, which is more suitable for

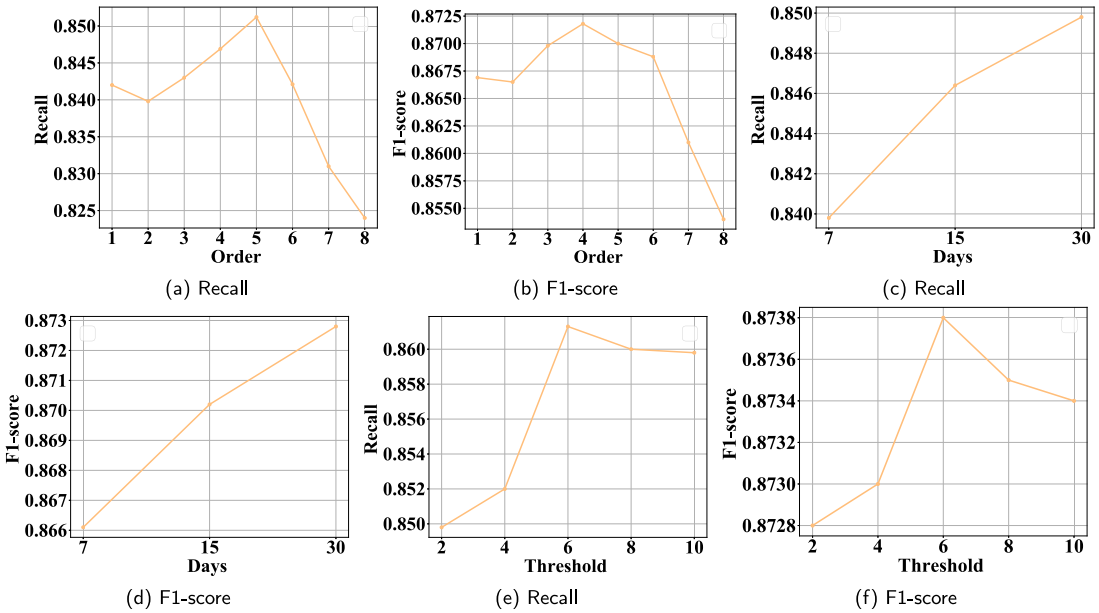


Fig. 6. The effects of key parameters.

Table 3
Ablation study results. F1 (abbr. F1-macro).

Tele-Max												
Method	10%			20%			30%			40%		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
CORE-DGNN-v1	0.6836	0.6690	0.6619	0.6998	0.6702	0.6700	0.7154	0.6781	0.6788	0.7213	0.6923	0.7122
CORE-DGNN-v2	0.7778	0.7477	0.7089	0.7898	0.7563	0.7388	0.7923	0.7611	0.7398	0.8044	0.7613	0.7467
CORE-DGNN-v3	0.8703	0.8197	0.8429	0.8709	0.8313	0.8587	0.8953	**0.8423**	0.8606	0.9046	0.8419	0.8672
CORE-DGNN-v4	0.8720	**0.8277**	0.8400	0.8891	0.8213	0.8443	0.8923	0.8231	0.8675	0.9092	0.8461	0.8708
CORE-DGNN	**0.9078**	0.8265	**0.8533**	**0.9103**	**0.8330**	**0.8591**	**0.9118**	0.8360	**0.8619**	**0.9129**	**0.8498**	**0.8728**
Tele-Min												
CORE-DGNN-v1	0.7230	0.6833	0.6932	0.7411	0.6922	0.7002	0.7745	0.7421	0.7299	0.7922	0.7399	0.7534
CORE-DGNN-v2	0.8018	0.7733	0.7521	0.7645	0.7433	0.7523	0.8121	0.7543	0.7591	0.8315	0.7823	0.7763
CORE-DGNN-v3	0.8813	0.8231	0.8222	0.8801	0.8203	0.8334	0.8998	0.8321	0.8406	0.9000	0.8309	0.8502
CORE-DGNN-v4	0.8922	0.8423	0.8301	0.9092	0.8344	0.8421	0.9100	0.8341	0.8521	0.9186	0.8477	0.8633
CORE-DGNN	**0.9110**	**0.8591**	**0.8423**	**0.9190**	**0.8410**	**0.8501**	**0.9292**	**0.8499**	**0.8634**	**0.9200**	**0.8599**	**0.8800**

modeling discontinuous frauds than the traditional RNN-based methods. (3) It further designs a frequency filtering method to alleviate the heterophily problem in collaborative networks.

5.5. Ablation study

Our model has four major components, i.e., collaborative network, Beta Wavelet graph neural network, temporal convolutional network, and self-attention mechanism. We conduct an ablation study by removing components respectively. (1) CORE-DGNN-v1: we remove the collaborative network and only mine the individual sequences pattern. (2) CORE-DGNN-v2: we replace the multi-frequency graph neural network with plain GNN, which does not address the heterophily problem in a collaborative network. (3) CORE-DGNN-v3: we replace the temporal convolutional network with a max-pooling module with RNN. (4) CORE-DGNN-v4: we remove the self-attention mechanic.

As shown in Table 3, we can easily see that:

- Without the collaborative network and Beta Wavelet graph neural network, CORE-DGNN-v1 and CORE-DGNN-v2 achieved the worse performance in the ablation study, indicating that mining collaborative fraud strategies and addressing the heterophily problem are key components for improving fraud detection performance. The collaborative network contributed more significantly to the performance of CORE-DGNN, indicating that collaborative fraud is already prevalent in telecom fraud.
- CORE-DGNN-v3 and CORE-DGNN-v4 achieved worse performance, indicating that temporal convolutional networks and self-attention mechanisms are crucial to CORE-DGNN. The performance of the CORE-DGNN-v4 was closest to that of the full model, suggesting that the self-attention mechanism contributes less to the success of the CORE-DGNN than the other three components.

5.6. Effects of parameters

We investigated the effects of key parameters on the performance of the CORE-DGNN method, as shown in Fig. 6. Owing to space limitations, we only present the Recall and F1-score when the train sets as 40% in Tele-max.

The Order C in Collaborative Pattern Mining. As shown in Fig. 6(a)(b), the performance of CORE-DGNN first improved and then deteriorated. CORE-DGNN achieved the best performance when the C was 5. The performance of CORE-DGNN improved as C increased from 1 to 5. This is easy to understand, as when the C value is too small, the filter has a smaller field of view, which leads to the aggregation of fewer neighbors with the same

label. However, when the C value increases beyond a certain level, the performance deteriorates. This may be because a larger filter field of view inevitably introduces more noise, which leads to lower classification accuracy. Therefore, we should determine an appropriate C. Considering the characteristics of the performance, the default value is set to 5.

The Resolution of Time Slice. As shown in Fig. 6(c)(d), the performance of the CORE-DGNN method improved as the interval of time slice increased from 7 days to 30 days. This may be because as victims become more aware of anti-fraud, the current collaborative fraud strategies used by fraudsters have shifted from short-time collaboration to long-time collaboration. Collaborative fraud at one-month (30 days) intervals is more likely to hide the characteristics of collaborative fraud to reduce the chance of being detected by victims than collaborative fraud at one-week (7 days) intervals. However, too large an interval can reduce the efficiency of the system response, and considering the performance and efficiency, the resolution of the time slice is set to 15 days.

The Threshold of Common Recipients. As shown in Fig. 6(e)(f), the performance of the CORE-DGNN method improves as the threshold of common recipients increases from 2 to 6. The performance of CORE-DGNN deteriorates when the threshold value is greater than 6. The explanations are as follows: (1) When the common recipient's threshold increases, it may be more accurate to establish edges between fraudsters, which reduces the network heterophily problem. (2) When the threshold is too large, the stricter setting incorrectly excludes some gang members and breaks the gang structure. Hence, we set the threshold as 6.

6. Conclusion and future work

In this paper, we study the problem of telecom fraud mining in a large-scale telecom network. By analyzing 5.0 million telecom records between 6106 users and 1.25 million recipients over eight months in a Chinese province, we find that current telecom fraud is dominated by collaborative fraud strategies. In addition, the telephone social behavior of fraudsters shows a dynamic evolution over time, while normal users remain relatively stable. Inspired by our exploratory analysis, we propose a time-slice-based dynamic graph neural network method to distinguish fraudsters from normal users. Experimental results show that our model achieves significant improvements compared with several state-of-the-art baseline approaches.

However, the current research focuses on fraud detection using the same distribution assumptions for the test and training sets, it is interesting to think about how to design a general fraud model with better generalization capabilities to address the problem of out-of-distribution in fraud detection. In addition, some studies [44,45] have confirmed that multi-modal fusion methods can achieve a more comprehensive description of data items. In the future, we can use multimodal data such as voice, video, and text for more accurate modeling of fraudulent behavior.

CRediT authorship contribution statement

Lingfei Ren: Conceptualization, Methodology, Software, Writing – original draft. **Ruimin Hu:** Funding acquisition, Resources, Supervision. **Dengshi Li:** Funding acquisition, Resources, Supervision. **Yang Liu:** Data curation, Validation. **Junhang Wu:** Data curation, Validation. **Yilong Zang:** Formal analysis, Investigation. **Wenyi Hu:** Formal analysis, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request

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